NITHISHVELAN M

Application Security & DevSecOps Engineer
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PROFESSIONAL SUMMARY

Application Security & Systems Engineer specializing in Static Application Security Testing (SAST), Abstract Syntax Tree (AST) analysis, security telemetry (OASIS SARIF 2.1.0), and automated vulnerability gating. Upstream open-source contributor to the Linux Kernel and security advisory databases. Experienced in developing custom security scanning engines, Shannon entropy credential detectors, and DevSecOps CI/CD pipelines.

CORE TECHNICAL SKILLS

- **Security & SAST:** Static Application Security Testing (SAST), AST Traversal, OASIS SARIF 2.1.0, Threat Modeling (STRIDE), Shannon Entropy Heuristics, OWASP Top 10, CWE Mapping, Zero-Leakage Data Masking.
- **Languages & Systems:** Python 3, C++17/20, C, Bash/POSIX Shell, SQL, Git Internals.
- **DevSecOps & Tooling:** GitHub Actions CI/CD, Docker, Pre-Commit Framework, Flake8, Pytest, Valgrind, Linux Security Modules.

UPSTREAM OPEN SOURCE SECURITY CONTRIBUTIONS

- **Linux Kernel (`drivers/staging/iio/ad5933.c`)** — *Kernel Concurrency & Lock Modernization*
- Refactored multi-branch error-path mutex handling in the AD5933 impedance analyzer driver to use RAII `guard(mutex)` ( ), structurally eliminating lock-leakage deadlocks.
- Formatted and submitted patches adhering strictly to LKML conventions via `git send-email` to upstream subsystem maintainers (Jonathan Cameron, Greg Kroah-Hartman).
- **GitHub Advisory Database (`github/advisory-database`)** — *Security Advisory Research*
- Audited and contributed vulnerability scope documentation improvements and metadata clarification for published CVE advisories.
- **boykopovar/AnyPS5** — *Static Analysis & Header Invariant Enforcement*
- Resolved duplicate symbol declaration failures detected by Python AST static analysis (`public_headers`), ensuring single-source-of-truth export hygiene across 12,000+ virtual ABI symbols.

FEATURED SECURITY PROJECTS

ast-guard — *AST-Based Static Application Security Testing Engine & GitHub Action*

- Engineered a lightweight Python AST SAST scanner analyzing code structures via `ast.NodeVisitor` with **0% comment/docstring false-positive rates** compared to regex linters.
- Emits native **OASIS SARIF 2.1.0** telemetry, enabling automated security alerts within GitHub Code Scanning and PR annotation gates.
- Benchmarked against 100,000+ LOC codebases; executes full security rule evaluation in **`< 1.2 seconds`** with `< 48MB` RSS memory overhead.
- Published as a reusable Composite **GitHub Action** (`ErenYeager1836/ast-guard@v1`) with configurable severity gates (`--fail-on HIGH`).
- Authored **ADR 0001: AST Visitor Traversal vs. Regular Expression Pattern Matching**.

secret-scanner — *High-Entropy Credential Detection CLI & Git History Auditor*

- Built an automated secret scanner implementing **Shannon entropy heuristics** ($H \ge 3.2$) alongside regex pattern matching to detect unpatterned API tokens and cryptographic keys.
- Implemented full Git DAG history inspection (`git log -p`), detecting leaked credentials retroactively deleted in subsequent commits.
- Enforced **guaranteed dynamic masking** to prevent cleartext secret leakage across stdout and JSON output artifacts.
- Benchmarked sustained filesystem scan throughput at **`~85 MB/s`** with `< 42MB` memory footprint.

reconflow — *Modular Security Reconnaissance CLI (Published on PyPI)*

- Developed a multi-threaded reconnaissance tool orchestrating passive DNS enumeration, subdomain discovery, and live HTTP probing.
- Selected and indexed by the **XERJ Corpus Hub** as an open-source reference implementation for AI coding agents.

security-labs — *Defensive Security & Vulnerability Simulation Suite*

- Developed socket-level port scanners, authentication log analyzers, and automated Linux host hardening auditing scripts.
- Configured an intentionally vulnerable multi-tier test lab in Docker demonstrating SQLi, IDOR, and privilege escalation defenses.

EDUCATION

- **Bachelor of Engineering (B.E.) in Materials Science and Engineering**
  College of Engineering, Guindy (CEG), Anna University, Chennai